

Advanced (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $x(x + 3) + 2(x + 3)$
- (B) $(x + 2)(x + 3)$
- (C) $(x + 1)(x + 6)$
- (D) $x(x + 2) + 3(x + 2)$
- (E) $(x + 4)(x + 2)$

Q2. Factor the trinomial $x^2 - 7x + 12$.

- (A) $(x - 3)(x - 4)$
- (B) $(x - 2)(x - 5)$
- (C) $(x - 1)(x - 6)$
- (D) $x(x - 3) + 4(x - 3)$
- (E) $(x - 4)(x - 3)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $(x + 5)(x - 3)$
- (B) $(x + 3)(x - 5)$
- (C) $(x + 2)(x - 7)$
- (D) $(x + 1)(x - 15)$
- (E) $(x - 2)(x + 7)$

Q4. Factor the trinomial $x^2 - 4x - 21$.

- (A) $(x - 7)(x + 3)$
- (B) $(x + 7)(x - 3)$
- (C) $(x - 2)(x - 10)$
- (D) $(x + 1)(x - 20)$
- (E) $(x - 3)(x + 7)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $(x + 4)(x + 6)$
- (B) $(x + 2)(x + 12)$
- (C) $(x + 1)(x + 23)$
- (D) $(x + 3)(x + 8)$
- (E) $(x + 6)(x + 4)$

Q6. Factor the trinomial $x^2 - 9x + 20$.

- (A) $(x - 4)(x - 5)$
- (B) $(x - 2)(x - 10)$
- (C) $(x - 1)(x - 19)$
- (D) $(x - 3)(x - 6)$
- (E) $(x - 5)(x - 4)$

Q7. Factor the trinomial $x^2 + 6x + 8$.

- (A) $(x + 2)(x + 4)$
- (B) $(x + 1)(x + 7)$
- (C) $(x + 3)(x + 3)$
- (D) $(x + 4)(x + 2)$
- (E) $(x + 5)(x + 1)$

Q8. Factor the trinomial $x^2 - 2x - 15$.

- (A) $(x - 5)(x + 3)$
- (B) $(x + 5)(x - 3)$
- (C) $(x - 2)(x - 7)$
- (D) $(x - 1)(x - 14)$
- (E) $(x + 3)(x - 5)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 4x - 21$. Show all work and provide the factored form.

Q10. Factor the trinomial $x^2 - 6x - 16$. Show all work and provide the factored form.

Chef's Tips

- Check for the correct factored form.
- Verify that students have applied the correct factoring method.
- Ensure students show all work for Free Response questions.